

ASTA Academy

Digital SAT Math Practice Set

Sample Set — 25 Questions

All four math content domains, with full answer key and step-by-step explanations.

Contents

Domain	Questions	Question IDs
Algebra	7	ALG
Advanced Math	6	ADV
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Geometry and Trigonometry	5	GEO

Practice Questions

Domain 1: Algebra

Linear equations, linear functions, systems of linear equations, and linear inequalities.

Question 1 [ALG-001] Difficulty: Easy

Skill: Linear equations in one variable

If $3(x - 4) = 2x + 5$, what is the value of x ?

- (A) -17
- (B) -7
- (C) 7
- (D) 17

Question 2 [ALG-002] Difficulty: Easy

Skill: Linear functions

The function f is defined by $f(x) = 5x - 8$. What is the value of $f(6)$?

- (A) 2
- (B) 11
- (C) 22
- (D) 30

Question 3 [ALG-003] Difficulty: Medium

Skill: Systems of two linear equations

If $2x + 3y = 19$ and $x - y = 2$, what is the value of $x + y$?

- (A) 5
- (B) 7
- (C) 8
- (D) 9

Question 4 [ALG-004] Difficulty: Medium

Skill: Linear inequalities

Which of the following values of x satisfies the inequality $-2x + 7 \geq 3$?

- (A) $x = 3$
- (B) $x = 2.5$

(C) $x = 2$

(D) $x = 5$

Question 5 [ALG-005] Difficulty: Medium

Skill: Linear equations in two variables

A line in the xy -plane passes through the points $(2, 5)$ and $(6, 13)$. What is the slope of the line?

(A) $\frac{1}{2}$

(B) 2

(C) 4

(D) 8

Question 6 [ALG-006] Difficulty: Medium

Skill: Linear word problems

A gym charges a one-time membership fee of \$40 plus \$25 per month. If Jordan paid a total of \$290, for how many months was Jordan a member?

Student-Produced Response — enter your answer:

Question 7 [ALG-007] Difficulty: Hard

Skill: Systems of linear equations (no solution / infinite)

The system of equations $4x + ky = 10$ and $8x + 6y = 20$ has infinitely many solutions. What is the value of k ?

(A) 1.5

(B) 2

(C) 3

(D) 6

Domain 2: Advanced Math

Equivalent expressions, nonlinear equations in one variable, systems of nonlinear equations, and nonlinear functions.

Question 8 [ADV-001] Difficulty: Easy

Skill: Equivalent expressions

Which expression is equivalent to $(2x + 3)(x - 5)$?

- (A) $2x^2 - 7x - 15$
- (B) $2x^2 + 7x - 15$
- (C) $2x^2 - 13x - 15$
- (D) $2x^2 - 7x + 15$

Question 9 [ADV-002] Difficulty: Medium

Skill: Quadratic equations

What are the solutions to the equation $x^2 - 6x + 8 = 0$?

- (A) $x = -2$ and $x = -4$
- (B) $x = 2$ and $x = 4$
- (C) $x = 1$ and $x = 8$
- (D) $x = -1$ and $x = -8$

Question 10 [ADV-003] Difficulty: Medium

Skill: Exponential functions

A bacterial population doubles every 3 hours. If the initial population is 500, which function $P(t)$ gives the population after t hours?

- (A) $P(t) = 500 \cdot 2^{(3t)}$
- (B) $P(t) = 500 \cdot 2^{(t/3)}$
- (C) $P(t) = 500 + 2t/3$
- (D) $P(t) = 500 \cdot 3^{(t/2)}$

Question 11 [ADV-004] Difficulty: Medium

Skill: Quadratic functions and vertex

The function f is defined by $f(x) = (x - 4)^2 - 9$. What is the minimum value of $f(x)$?

- (A) -9
- (B) -4

- (C) 4
- (D) 9

Question 12 [ADV-005] Difficulty: Hard

Skill: Radical and rational equations

If $\sqrt{3x - 2} = x - 2$, what is the value of x ?

- (A) 1
- (B) 3
- (C) 6
- (D) Both 1 and 6

Question 13 [ADV-006] Difficulty: Hard

Skill: Polynomial factors and zeros

The polynomial $p(x) = x^3 - 4x^2 + x + 6$ has $(x + 1)$ as a factor. What are all the zeros of $p(x)$?

- (A) -1, 2, 3
- (B) -1, -2, 3
- (C) 1, 2, 3
- (D) -1, 2, -3

Domain 3: Problem-Solving and Data Analysis

Ratios, rates, proportional relationships, percentages, units, statistics, and probability.

Question 14 [PSD-001] Difficulty: Easy

Skill: Percentages

A jacket originally priced at \$80 is on sale for 25% off. What is the sale price?

- (A) \$20
- (B) \$55
- (C) \$60
- (D) \$65

Question 15 [PSD-002] Difficulty: Easy

Skill: Ratios and proportions

The ratio of red marbles to blue marbles in a jar is 3:5. If there are 24 red marbles, how many blue marbles are there?

- (A) 15
- (B) 32
- (C) 40
- (D) 45

Question 16 [PSD-003] Difficulty: Medium

Skill: Unit rates

A car travels 252 miles on 9 gallons of gasoline. At this rate, how many gallons are needed to travel 420 miles?

- (A) 12
- (B) 14
- (C) 15
- (D) 16

Question 17 [PSD-004] Difficulty: Medium

Skill: Percentage change

The population of a town grew from 12,500 to 14,000 over one year. What was the approximate percent increase?

- (A) 10%

- (B) 12%
- (C) 15%
- (D) 20%

Question 18 [PSD-005] *Difficulty: Medium*

Skill: Mean, median, range

The data set {4, 7, 9, 12, 8, 10} has mean m . If a new value of 14 is added to the data set, what happens to the mean?

- (A) It decreases
- (B) It stays the same
- (C) It increases
- (D) Cannot be determined

Question 19 [PSD-006] *Difficulty: Medium*

Skill: Probability

A bag contains 4 red, 6 blue, and 10 green marbles. If one marble is drawn at random, what is the probability that it is NOT green?

- (A) $\frac{1}{4}$
- (B) $\frac{1}{2}$
- (C) $\frac{2}{5}$
- (D) $\frac{3}{5}$

Question 20 [PSD-007] *Difficulty: Hard*

Skill: Two-way tables

A survey of 200 students asked whether they prefer math or science. 70% of the students were freshmen, and 40% of freshmen preferred math. How many freshmen preferred math?

Student-Produced Response — enter your answer:

Domain 4: Geometry and Trigonometry

Area, volume, lines, angles, triangles, right triangles, trigonometric ratios, and circles.

Question 21 [GEO-001] Difficulty: Easy

Skill: Area and perimeter

A rectangle has a length of 12 cm and a width of 5 cm. What is the area of the rectangle, in square centimeters?

- (A) 17
- (B) 34
- (C) 60
- (D) 120

Question 22 [GEO-002] Difficulty: Easy

Skill: Pythagorean theorem

In a right triangle, the two legs measure 6 and 8. What is the length of the hypotenuse?

- (A) 10
- (B) 12
- (C) 14
- (D) $\sqrt{14}$

Question 23 [GEO-003] Difficulty: Medium

Skill: Circles

A circle has a radius of 7. What is the area of the circle?

- (A) 14π
- (B) 28π
- (C) 49π
- (D) 98π

Question 24 [GEO-004] Difficulty: Medium

Skill: Volume

A cylinder has a radius of 3 inches and a height of 10 inches. What is the volume, in cubic inches?

- (A) 30π
- (B) 60π
- (C) 90π

(D) 100π

Question 25 [GEO-005] *Difficulty: Hard*

Skill: Right triangle trigonometry

In a right triangle, $\sin(\theta) = 3/5$, where θ is one of the acute angles. What is the value of $\cos(\theta)$?

(A) $3/4$

(B) $4/5$

(C) $5/4$

(D) $5/3$

Answer Key

#	ID	Domain	Difficulty	Answer
1	ALG-001	Algebra	Easy	D
2	ALG-002	Algebra	Easy	C
3	ALG-003	Algebra	Medium	C
4	ALG-004	Algebra	Medium	C
5	ALG-005	Algebra	Medium	B
6	ALG-006	Algebra	Medium	10
7	ALG-007	Algebra	Hard	C
8	ADV-001	Advanced Math	Easy	A
9	ADV-002	Advanced Math	Medium	B
10	ADV-003	Advanced Math	Medium	B
11	ADV-004	Advanced Math	Medium	A
12	ADV-005	Advanced Math	Hard	C
13	ADV-006	Advanced Math	Hard	A
14	PSD-001	Problem-Solving and Data Analysis	Easy	C
15	PSD-002	Problem-Solving and Data Analysis	Easy	C
16	PSD-003	Problem-Solving and Data Analysis	Medium	C
17	PSD-004	Problem-Solving and Data Analysis	Medium	B
18	PSD-005	Problem-Solving and Data Analysis	Medium	C
19	PSD-006	Problem-Solving and Data Analysis	Medium	B
20	PSD-007	Problem-Solving and Data Analysis	Hard	56
21	GEO-001	Geometry and Trigonometry	Easy	C
22	GEO-002	Geometry and Trigonometry	Easy	A
23	GEO-003	Geometry and Trigonometry	Medium	C
24	GEO-004	Geometry and Trigonometry	Medium	C
25	GEO-005	Geometry and Trigonometry	Hard	B

Explanations

Algebra

Question 1 [ALG-001] — Answer: **D**

Distribute the 3 on the left: $3x - 12 = 2x + 5$. Subtract $2x$ from both sides: $x - 12 = 5$. Add 12 to both sides: $x = 17$. The answer is D.

Question 2 [ALG-002] — Answer: **C**

Substitute 6 for x : $f(6) = 5(6) - 8 = 30 - 8 = 22$. The answer is C.

Question 3 [ALG-003] — Answer: **C**

From the second equation, $x = y + 2$. Substitute into the first: $2(y + 2) + 3y = 19$, which gives $5y + 4 = 19$, so $5y = 15$ and $y = 3$. Then $x = 5$. Therefore $x + y = 5 + 3 = 8$. The answer is C.

Question 4 [ALG-004] — Answer: **C**

Solve: $-2x + 7 \geq 3 \rightarrow -2x \geq -4 \rightarrow x \leq 2$ (flip inequality when dividing by a negative). Only $x = 2$ satisfies $x \leq 2$. The answer is C.

Question 5 [ALG-005] — Answer: **B**

Slope = $(y_2 - y_1) / (x_2 - x_1) = (13 - 5) / (6 - 2) = 8/4 = 2$. The answer is B.

Question 6 [ALG-006] — Answer: **10**

Set up the equation: $40 + 25m = 290$, where m is the number of months. Subtract 40: $25m = 250$. Divide by 25: $m = 10$. Jordan was a member for 10 months.

Question 7 [ALG-007] — Answer: **C**

For infinitely many solutions, one equation must be a multiple of the other. The second equation is twice the first if $8x = 2(4x)$ ✓ and $20 = 2(10)$ ✓, so we need $6y = 2(ky)$, meaning $6 = 2k$, so $k = 3$. The answer is C.

Advanced Math

Question 8 [ADV-001] — Answer: **A**

FOIL: $(2x)(x) + (2x)(-5) + (3)(x) + (3)(-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$. The answer is A.

Question 9 [ADV-002] — Answer: **B**

Factor: $x^2 - 6x + 8 = (x - 2)(x - 4) = 0$. So $x - 2 = 0$ or $x - 4 = 0$, giving $x = 2$ or $x = 4$. The answer is B.

Question 10 [ADV-003] — Answer: **B**

Doubling every 3 hours means the exponent of 2 should equal 1 when $t = 3$. The form $P(t) = 500 \cdot 2^{(t/3)}$ gives $P(3) = 500 \cdot 2 = 1000$ ✓. The answer is B.

Question 11 [ADV-004] — Answer: **A**

The function is in vertex form $f(x) = (x - h)^2 + k$, where the vertex is at $(h, k) = (4, -9)$. Since the coefficient of the squared term is positive, the parabola opens upward, so the vertex is the minimum. The minimum value is -9 . The answer is A.

Question 12 [ADV-005] — Answer: **C**

Square both sides: $3x - 2 = (x - 2)^2 = x^2 - 4x + 4$. Rearrange: $x^2 - 7x + 6 = 0$. Factor: $(x - 1)(x - 6) = 0$, so $x = 1$ or $x = 6$. Always check radical equations for extraneous solutions. For $x = 1$: $\sqrt{(3 \cdot 1 - 2)} = \sqrt{1} = 1$, but $x - 2 = -1$, so $1 \neq -1$ (extraneous). For $x = 6$: $\sqrt{(3 \cdot 6 - 2)} = \sqrt{16} = 4$, and $x - 2 = 4$ ✓. The only valid solution is $x = 6$. The answer is C.

Question 13 [ADV-006] — Answer: **A**

Since $(x + 1)$ is a factor, divide $p(x)$ by $(x + 1)$ using synthetic or polynomial division. The result is $x^2 - 5x + 6$, which factors as $(x - 2)(x - 3)$. So $p(x) = (x + 1)(x - 2)(x - 3)$, giving zeros at $x = -1, 2$, and 3 . The answer is A.

Problem-Solving and Data Analysis

Question 14 [PSD-001] — Answer: **C**

25% of $\$80 = 0.25 \times 80 = \20 discount. Sale price = $\$80 - \$20 = \$60$. The answer is C.

Question 15 [PSD-002] — Answer: **C**

Set up the proportion: $3/5 = 24/x$. Cross-multiply: $3x = 120$, so $x = 40$. The answer is C.

Question 16 [PSD-003] — Answer: **C**

Miles per gallon: $252 \div 9 = 28$ mpg. Gallons needed for 420 miles: $420 \div 28 = 15$. The answer is C.

Question 17 [PSD-004] — Answer: **B**

Increase = $14,000 - 12,500 = 1,500$. Percent increase = $(1,500 / 12,500) \times 100 = 12\%$. The answer is B.

Question 18 [PSD-005] — Answer: **C**

Original mean = $(4+7+9+12+8+10)/6 = 50/6 \approx 8.33$. New mean = $64/7 \approx 9.14$. Since the new value (14) is greater than the original mean, adding it increases the mean. The answer is C.

Question 19 [PSD-006] — Answer: **B**

Total marbles = $4 + 6 + 10 = 20$. Non-green = $4 + 6 = 10$. Probability = $10/20 = 1/2$. The answer is B.

Question 20 [PSD-007] — Answer: **56**

Number of freshmen = 70% of 200 = 140. Freshmen who prefer math = 40% of 140 = $0.4 \times 140 = 56$.

Geometry and Trigonometry

Question 21 [GEO-001] — Answer: **C**

Area of rectangle = length $\times$ width = $12 \times 5 = 60$ square cm. The answer is C.

Question 22 [GEO-002] — Answer: **A**

By the Pythagorean theorem: $c^2 = 6^2 + 8^2 = 36 + 64 = 100$, so $c = 10$. The answer is A.

Question 23 [GEO-003] — Answer: **C**

Area of circle = $\pi r^2 = \pi(7)^2 = 49\pi$. The answer is C.

Question 24 [GEO-004] — Answer: **C**

Volume of cylinder = $\pi r^2 h = \pi(3)^2(10) = 90\pi$. The answer is C.

Question 25 [GEO-005] — Answer: **B**

If $\sin(\theta) = \text{opposite/hypotenuse} = 3/5$, then opposite = 3 and hypotenuse = 5. By Pythagorean theorem: $\text{adjacent}^2 = 5^2 - 3^2 = 25 - 9 = 16$, so adjacent = 4. Therefore $\cos(\theta) = \text{adjacent/hypotenuse} = 4/5$. The answer is B.